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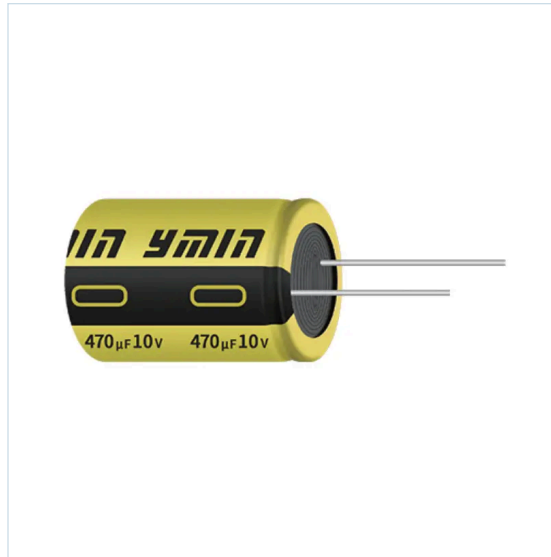


LKX



LKM

LKM



Short Description:

Aluminum Electrolytic Capacitor

Radial Lead Type

Small size, high frequency and large ripple current resistance,
high frequency and low impedance, power supply products,
7000~10000 hours in 105°C environment, comply with AEC-Q200 RoHS direct

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Main Technical Parameters

Technical Parameter

- ◆ 105°C 7000~10000 Hours
- ◆ Miniature, High Ripple Current, High Frequency, Low ESR
- ◆ RoHS Compliant
- ◆ AEC-Q200 Qualified, Please Consult Us For More Details

Specification

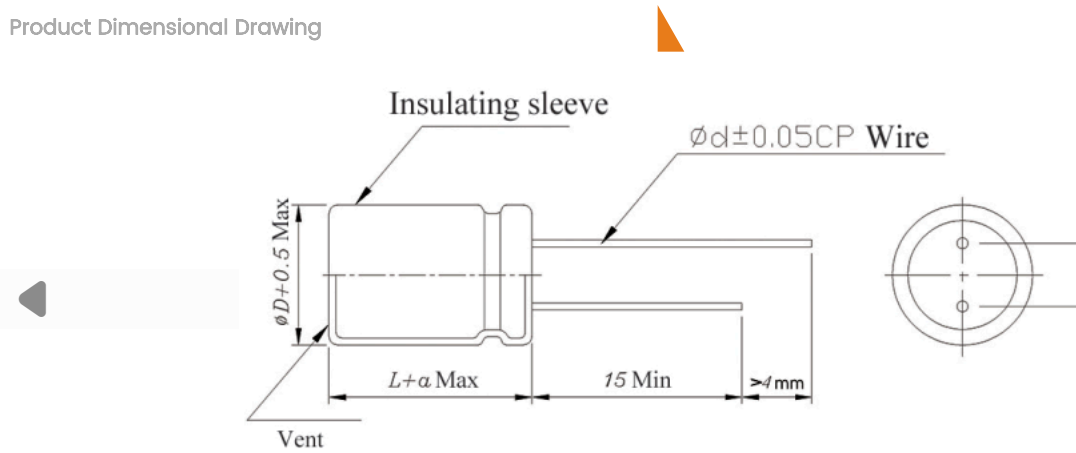
Items	Characteristics
Operation Temperature Range	≤120V.DC -55°C~+105°C ; 160~500V.DC -40°C~+105°C
Rated Voltage	10~500V.DC
Capacitance Tolerance	±20%(25±2°C 120Hz)
Leakage Current((uA)	10— 120WV ≤ 0.01CV or 3uA whichever is greater C:rated capacitance(uF) V:rated volt 160 ~500WV ≤ 0.02CV+10(uA) C:rated

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Dissipation Factor (25±2°C 120Hz)	Rated Voltage(V)	120							
	tgδ	0.09							
	For those with rated capacitance large increased by 0.02								
Temperature Characteristics (120Hz)	Rated Voltage(V)	10							
	Z(-40°C)/Z(20°C)	6							
	Rated Voltage(V)	120							
	Z(-40°C)/Z(20°C)	5	5	5	5	7	7		
Endurance	After standard test time with applying the rated voltage with the rated ripple current in the oven specification shall be satisfied after 16 hours at 25±2°C.								
	Capacitance change		within±20% of the initial value						
	Dissipation Factor		Not more than 200% of the specified value						
	Leakage current		Not more than the specified value						
	Load life(hours)	Φ5		70					
		Φ6.3		90					
≥Φ8		10							
Shelf Life At High Temperature	After leaving capacitors under no load at 105°C for 1000 hours, the following specification shall be								
	Capacitance change		within±20% of the initial value						
	Dissipation Factor		Not more than 200% of the specified value						
	Leakage current		Not more than 200% of the specified value						

Send

Product Dimensional Drawing



Remark: capacitors with diameter more than 6.3 have safety vent

Dimension (Unit: mm)

Remark: capacitors with diameter more than 6.3 have safety vent

top

L>16	$\alpha=2.0$
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D	5	6.3	8	10	12.5	14.5	11
d	0.5	0.5	0.6	0.6	0.6	0.8	0
F	2	2.5	3.5	5	5	7.5	7

Ripple Current Frequency Correction Coefficient

Ripple Current Correction Factor

Frequency Correction Factor

Frequency (Hz)	50	120	1K	10K-50K
Coefficient	0.4	0.5	0.8	0.9

Temperature Correction Factor

Environment Temperature(°C)	50°C	70°C	85°C
Correction Factor	2.1	1.8	1.4

LKM Series: Small Size, High Energy – Providing Stable Core Power for Modern High-Performance Electronic Devices

In today's era driven by electronic technology, the performance boundaries of electronic devices are constantly being pushed increasingly compact and complex. Whether it's speeding electric vehicles, constantly running data servers, or the increasing place unprecedentedly stringent demands on one of the most fundamental components – capacitors: they must maintain performance under extreme temperatures, severe voltage fluctuations, and high-frequency ripple currents. YMIN's LKM series aluminum electrolytic capacitors have become an indispensable and stable "heart" in many high-end electronic system designs.

I. Precise Positioning: A Solution for High Performance and High Reliability

The LKM series is not an ordinary general-purpose aluminum electrolytic capacitor, but a radially led product specifically designed for high performance. Its operating temperature range covers a wide range of -55°C to +105°C, ensuring stable operation, especially at 105°C, with a life expectancy that even under severe conditions of internal heat accumulation, LKM capacitors can operate continuously for nearly a year with performance degradation remaining within standard limits. More importantly, the entire series complies with the AEC-Q200 standards, the automotive electronics industry's top requirements for component reliability, environmental adaptability, and long lifespan. From Advanced Driver Assistance Systems (ADAS) to the electronic control units of new energy vehicles, the LKM series provides the power sought by automotive electronics.

II. Core Technological Advantages: Analyzing the Deeper Meaning of "High Performance"

The product description of the LKM series contains several key technological advantages, which together form the core of its high performance.

and GaN fast chargers), it can more effectively filter high-frequency noise, smooth the output voltage, and provide cleaner, r CPUs and GPUs, while generating less heat itself, thus improving overall system efficiency.

2. High Ripple Current Withstand Capability: Ripple current is a major factor causing capacitor heating and affecting its lifespan. For currents far exceeding those of ordinary capacitors, thanks to its lower internal ESR and optimized heat dissipation design. For capacitor can withstand a ripple current of up to 1.8A at 105°C. This characteristic is crucial for applications handling instantaneous in motor drives, uninterruptible power supplies (UPS), and industrial power supplies, ensuring system stability and the longev drastic load changes.

3. Superior Lifespan and Stability: The 7000-10000-hour lifespan promised by the LKM series is not an exaggeration, but rather tests require that after high-temperature full-load testing, the capacitor's capacitance change, loss tangent ($\tan\delta$), and leakage specified stringent range. This excellent stability reduces the risk of performance degradation or failure due to capacitor aging equipment requiring continuous operation year-round (such as communication base stations, smart meters, and energy storage

4. Robust Physical Structure and Safety: For capacitors with a diameter greater than 6.3mm, the LKM series is designed with a mechanism can release pressure in a controlled manner when the capacitor experiences a surge in internal pressure due to voltage surges or circuit errors), preventing the casing from rupturing and thus avoiding secondary damage to surrounding entire device.

III. Wide Range of Application Scenarios: Empowering Electronic Innovation Across Industries

The technical characteristics of the LKM series make it shine in several cutting-edge technology fields:

- **Automotive Electronics:** With the wave of automotive electrification and intelligentization, the number of in-vehicle electronic certification makes it an ideal choice for key components such as engine control units (ECUs), battery management system, LED headlight drivers, ensuring flawless operation in harsh environments such as vibration, high and low temperature shocks
- **AI Data Servers and Communication Equipment:** Data center server power supplies require extremely high power density and current capabilities effectively meet the stringent requirements of CPU/GPU power supply modules (VRMs) for decoupling and AI computing clusters and cloud computing infrastructure. Similarly, in 5G communication equipment, it can also handle power
- **Robotics and Industrial Automation:** Servo drives in industrial robots and CNC machine tools need to handle high-frequency capacitors, in the inverters and DC-Link circuits of such devices, can effectively smooth the DC bus voltage, absorb regeneration and system response speed.
- **PD Fast Charging and Smart Lighting:** PD fast charging heads based on gallium nitride (GaN) technology pursue extreme mini size (minimum diameter 5mm) and high-frequency, low-impedance characteristics of LKM perfectly match the high switch-mode charging circuits, making it a key component for achieving high-power, small-size fast charging. In the field of smart LED lighting power supply and stable light output.
- **New Energy and Smart Grid:** In photovoltaic inverters, wind power converters, and smart meters, LKM capacitors play a crucial filtering. Their long lifespan and high-temperature stability directly impact the power generation efficiency and operation and system.

IV. Choosing the LKM Series: Choosing Reliability and Superior Performance

In summary, the YMIN LKM series aluminum electrolytic capacitors are not merely simple electronic components; they are high materials science, precision manufacturing processes, and rigorous quality control. They represent the forefront of modern electronics designed to address the core pain points faced by engineers designing next-generation high-performance, high-reliability electronics.

When you choose the LKM series for your product, you are choosing a commitment to stability, a guarantee of long lifespan, and your product stand out in a highly competitive market. Whether it's meeting stringent automotive electronics standards, satisfying data centers, or achieving miniaturization and high performance in consumer electronics, the LKM series will be your trusted foundation for your innovative ideas.

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